FREQUENTLY ASKED QUESTIONS

Colorado Public Lands Mobile Location Data Dashboard

A guide to reading, interpreting, and responsibly using the statewide recreation visitation dashboard at:

PREPARED FOR

Colorado Parks and Wildlife

FUNDED BY

Great Outdoors Colorado

PREPARED BY

RRC Associates and SE Group

How to use this guide

The mobile location data dashboard uses anonymous mobile device data to show how people use Colorado's public lands. It reveals statewide patterns that were previously difficult to measure, but like any estimate, it should be interpreted carefully.

Where applicable, questions below include two levels of detail in their response: 1) a brief answer with key points, followed by 2) a "More Detail" note with added context, caveats, and examples. If you are new to the dashboard, start with three sections before drawing conclusions: Understanding the Metrics, Interpreting the Results, and Data Privacy and Anonymization.

Executive Summary

The Colorado Public Lands Mobile Location Data Dashboard is a free public tool that estimates patterns of recreation use on public lands across Colorado for calendar years 2024 and 2025. Funded by Great Outdoors Colorado (GOCO) and developed for Colorado Parks and Wildlife (CPW) by RRC Associates with SE Group, it uses anonymous mobile device location data to give the first statewide, consistently measured picture of recreation use.

The dashboard is designed for comparisons first and foremost. It is well suited to questions about which places see more relative use, when use peaks, and where visitors come from. It should not be used as a complete picture of total visitation or what activity users were engaged in. The data is presented as a sample of devices collected on-site.

Privacy and anonymity

Protecting individual privacy is an important component of this project.

- **Anonymized before the project team receives them.** Location data are licensed from Azira and arrive anonymized. Device identifiers are cryptographically hashed before delivery, and the data comply with privacy regulations including California's CCPA and the EU's GDPR.
- **No direct identifiers.** RRC, CPW, and project partners never receive names, phone numbers, email addresses, or account information. There is no step in the project workflow at which identifiable data are present.
- **No one is tracked for visiting.** Devices enter the dataset through consented data sharing tied to mobile apps and advertising. No one is tracked because they visited a park, and there is no way to opt into the study. Device owners control participation through the privacy settings on their own phones and apps.
- **Only aggregate results are displayed.** Every figure in the dashboard is a total across many devices. There is no way to select an individual device, view a single device's path, or drill down to a person.
- **Small samples are suppressed.** Results based on fewer than 30 device days are suppressed, because in a small enough group a pattern can point to an individual. This is standard practice in Census and public health reporting, and it means remote, lightly used places are the most likely to appear blank.
- **Home locations and demographics are area-based.** The data provider infers a likely home area from where a device consistently spends the night over roughly three months, then adds random jitter so it cannot be tied to a household. The finest geography ever available to the project is the Census Block Group. Demographics are never collected from individuals. They are American Community Survey characteristics of that area, describing the communities visitors come from rather than the visitors themselves.

How to use the dashboard

- **Select a location.** Search by name, or narrow the list using the CPW Region, County, Ownership, Management, RPI (Regional Partnership Initiative), and Custom List filters. A parcel may appear under more than one geography if any part of it intersects.
- **Apply your desired filters (Year, Season, and Roads filters).** Year shows 2024, 2025, or both. Season groups months. Roads includes or removes activity along road corridors, much of which is travel rather than recreation.
- **Compare like with like.** Hold the location, season, metric, and Roads setting constant when comparing, and check the Area figure first, since larger parcels record more activity simply because they are larger.
- **Export and share.** The download icon in the lower right saves images or PDFs (set page size to "unspecified") and the share icon copies a link that preserves your selections.
- **Expect some load time.** Large selections take time to load, and the dashboard is easiest to read on a full-size desktop or laptop screen.

Definitions of the major metrics

Six measures appear in the dashboard. They answer different questions and are not interchangeable.

Measures	Definition
Unique Devices	Distinct mobile devices observed at the location during the selected period. A device that returns many times is counted once. Describes breadth of use.
Device Days	One device observed at the location on one day. A device seen on five days contributes five device days. Describes frequency of use.
Visits	Discrete trips to the location. A device may record several visits in a day if it exits and re-enters the boundary or is absent two hours and returns. A device must be present at least five minutes to qualify.
Observations	Total location signals, or pings, recorded at the location. Not capped per device per day, so a device that lingers generates many. Describes intensity or duration.
Weighted Count	A modeled estimate of relative visitation, not a raw count. The sample is expanded using Census population data and other real-world counts to approximate total use.
Area (sq. mi.)	The size of the selected location.

The first four measures in this table are exact counts of what appeared in the sample. Weighted Count is a modeled estimate.

Interpreting results responsibly

- **These are estimates, not counts.** The dataset captures roughly 5 to 10 percent of devices present, and one device does not reliably equal one person. No figure in the dashboard should be perceived as the number of people who visited.
- **Relative patterns are the point.** Sampling limitations largely cancel out in comparison. If a similar share of visitors is captured at two sites, the ratio between them is informative even though neither total is correct.
- **Coverage varies.** Cell coverage, GPS accuracy, device settings, and how often apps report all shift results in ways unrelated to visitation, and they work against remote areas.
- **Blank does not mean empty.** A suppressed or zero result means too little qualifying activity was recorded, not that the location was empty.

- **Treat small differences as noise.** Two years show change but do not establish a trend. A five percent difference is within sampling noise; a forty percent shift is worth investigating, alongside closures, fires, weather, events, or new access.
- **Use it alongside other sources.** The dashboard shows what happened, rarely why. Pair it with counters, surveys, and local knowledge. Where it contradicts what site staff observe, the local observation is usually right. Questions, suspected errors, and requests for additional analysis: movement@rrcassociates.com

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1. About the Dashboard

This section explains what the dashboard is, who built it, and what it can and cannot tell you.

What is the CPW Mobile Location Data Dashboard?

It is a free, public, online tool that shows patterns of visitation to public lands across Colorado, estimated from anonymous mobile device location data. You can explore how use varies by place, by year, and by season.

> **MORE DETAIL** The dashboard is available at www.rrcmovement.com/copubliclandsdashboard. A link can also be found on the Colorado's Outdoors Strategy Resource Hub. It uses anonymous signals from mobile devices to estimate how many devices were present on a public land parcel, for how long, where they reside, and several other metrics. Because the underlying data represent the entire state, the dashboard enables insights into recreation patterns on lands that have not had visitor research conducted, including remote trailheads, State Wildlife Areas, BLM parcels, and similar public land locations.

What are the study's research goals?

The project set out to build the first statewide picture of recreation use on Colorado's public lands, using a consistent methodology everywhere, including places with no staff or research history.

> **MORE DETAIL** The goals were to: (1) develop a repeatable, statewide approach to estimating relative recreation use; (2) make those estimates available to land managers, partners, and the public in an intuitive format (3) fill gaps where traditional visitor counting is not feasible; and (4) support regional and statewide conservation and recreation planning with a common dataset.

Who developed and funded the project?

Great Outdoors Colorado (GOCO) provided funding to Colorado Parks and Wildlife (CPW). RRC Associates developed and led the project with SE Group as a subconsultant.

What geographic areas and time periods are covered?

The dashboard covers public lands throughout Colorado for calendar years 2024 and 2025.

> **MORE DETAIL** Study locations include public lands statewide: State Parks, State Wildlife Areas, U.S. Forest Service and BLM lands, National Park Service units, county and municipal open space, and other publicly accessible parcels, as identified in the underlying public lands datasets.

Who is the intended audience?

Anyone with an interest in how Colorado's public lands are used, including land managers, planners, regional partnerships, local governments, conservation organizations, researchers, and members of the public.

> **MORE DETAIL** The dashboard was designed primarily to support land management and planning decisions, but it is public because the patterns it shows are of broad interest.

What planning and management questions can the dashboard help address?

It is best at comparative and directional questions such as which places see more relative use than others, when use peaks, visitor origins, and areas of visitor use.

> **MORE DETAIL** Examples of questions the dashboard can help with: Which sites in a region appear to receive the most use? How does summer use compare to shoulder season use at a given location? Where are visitors coming from? Which lightly studied areas may warrant closer attention, monitoring, or investment?

What questions is the dashboard not designed to answer?

The dashboard does not provide a precise count of visitation, tell you what people were doing, or describe use at a very small scale or a very short time window.

> **MORE DETAIL** The dashboard is not a substitute for a gate count, a permit system, or an on-site survey. It cannot tell you the activity someone was engaged in (hiking versus hunting versus fishing), how satisfied visitors were, how much they spent, or whether a specific trail segment is over capacity on a specific afternoon. It should not be used to set enforcement actions, to allocate funding on a per-visitor basis, or to make claims about precise visitor numbers at a single site.

2. Getting Started

How do I select a study location?

Use the selection panel to pick a location by name or narrow the list using the geographic and category filters. Once a location is selected, the metrics and charts update to reflect it. Locations may be searched using the search bar located on the right side of the panel.

How do the CPW Region, County, Ownership, Management, RPI, and Custom List options work?

These options provide ways to narrow the list of locations. Region, County, and RPI group locations geographically. A parcel may appear in more than one geography if any part of it intersects a region, county, or RPI. You can also filter locations by land ownership, management responsibility, or curated “Custom Lists” that highlight selected topics of interest.

MORE DETAIL **Search**: enter part of a location name to find it quickly. **CPW Region**: filters locations by CPW administrative region. **County**: filters locations by Colorado county. **Ownership**: filters by the entity that owns the land, such as a federal, state, county, municipal, or special district owner. **Management**: filters by the agency or organization responsible for day-to-day management, which may differ from the owner. **RPI**: filters locations by Regional Partnership Initiative area. **Custom List**: shows preselected groups of locations related to specific topics. These lists are intended as helpful starting points and are not exhaustive.

What is an RPI region?

RPI stands for Regional Partnership Initiative. Colorado’s regional partnerships are collaborative groups that plan for conservation, recreation, and public land management across a shared area, supported by CPW and the state. Filtering by RPI lets a partnership look at all the public lands where at least a portion of the parcel is within its own boundaries.

How do the Year, Season, and Roads filters work?

Use the Year filter to view 2024, 2025, or both years combined. Use Season to focus on a defined group of months. Use Roads to include or remove activity associated with roadways.

> **MORE DETAIL** **Year**: select 2024, 2025, or both. **Season**: seasons are defined by the months typically associated with each. Winter = Dec through Feb, Spring = Mar through May, Summer = Jun through Aug, Fall = Sep through Nov. **Roads**: much of the location data recorded on public lands comes from people driving on highways and major roads rather than recreating. The Roads filter lets you view results with some or all of that road-associated activity removed.

Why might a selection take time to load?

The underlying dataset is very large. Selections require the dashboard to process a lot of records before it can display the result.

Can I download figures or data?

Users can download dashboard images or PDFs by clicking the download icon in the lower-right corner of the dashboard. PDF exports work best when the page size is set to "unspecified." Users may also capture screenshots of specific dashboard views.

Underlying data cannot be downloaded.

> **MORE DETAIL** The mobile location data underlying the dashboard are licensed for this project and cannot be redistributed. You are welcome to use images and screenshots for presentations and reports, and we ask that you credit the dashboard and the project partners. If you need results in another form, or need an analysis the dashboard does not support, contact the project team using the address in the last section of this FAQ.

How do I share a version of the dashboard with my selections saved?

To share a specific dashboard view, click the share icon in the lower-right corner of the dashboard, then copy the generated link.

3. Understanding the Metrics

The dashboard reports several different measures of activity. They answer different questions, and they are not interchangeable.

Metric definitions

Six measures appear in the dashboard: Unique Devices, Device Days, Visits, Observations, Weighted Count, and Area.

> **MORE DETAIL** **Unique Devices**: the number of distinct mobile devices observed in the study location during the selected time period. A device that returns many times is still counted once.
>
> **Device Days**: one device observed in the study location on one day counts as one device day. A device seen on five different days contributes five device days.
>
> **Visits**: a count of discrete trips to the study location. Users may record multiple visits within a day if they are observed entering and exiting the parcel boundary or, if they are not observed within the parcel boundary for two hours, they are inferred to have left and a new visit is registered upon re-entry.
>
> **Observations**: the total number of individual location signals, or "pings," recorded within the study location. Unlike Device Days, observations are not capped at one per device per day, so a device that lingers generates many observations.

Weighted Count: a modeled estimate of relative visitation. The raw sample is expanded using Census data and other real-world count data to approximate the scale of total use.

Area (square miles): the size of the selected study location. Useful for putting activity counts in context, since larger areas will generally record more activity than smaller ones.

How are these metrics different?

Think of them as answering four different questions: how many devices, how many device days of presence, how many visits, and how much signal activity.

MORE DETAIL Here is a simple example. Suppose one device visits a park on Saturday and again on Sunday, and generates 40 pings each day. That device contributes 1 Unique Device, 2 Device Days, 2 Visits, and 80 Observations. Unique Devices tells you about breadth of use. Device Days tells you about frequency of use. Visits tells you whether a location often receives multiple visits within a day. Observations tells you about intensity or duration. Weighted Count is different in kind, because it is not a raw count at all but a model-based estimate of overall visitation scale.

Which metrics describe the mobile-data sample, and which are modeled estimates?

Unique Devices, Device Days, Visits, and Observations are counts of what was observed in the sample. Weighted Count is a modeled estimate. Area is a fixed geographic measurement.

MORE DETAIL This distinction matters when you interpret results. The sample-based counts are exact counts of the devices in our dataset, but that dataset is only a fraction of all visitors, so the counts are far lower than true visitation. The Weighted Count attempts to scale up to a fuller picture, but it introduces modeling assumptions and should be treated as an estimate with meaningful uncertainty. For most comparisons, either type works, as long as you compare like with like.

Why might the number of devices, device days, and visits differ substantially?

Because they measure different things. For example, a location with a small number of devices returning frequently will show far more device days than unique devices.

MORE DETAIL A neighborhood open space used daily by nearby residents may show relatively few unique devices but a very high number of device days. A destination site drawing one-time visitors from across the country may show the opposite: many unique devices, each contributing only one or two device days. Neither pattern is better or worse, but they describe places with differing use, and comparing them on a single metric may be misleading. Additionally, cell service can have an impact on some level of data volume.

Does one device represent one person?

Generally, we tend to keep devices specific to the unit and not associate with a person. While there is a high likelihood that one device equals one person, that assumption does not always hold.

MORE DETAIL Several things may break the one-to-one assumption. Not everyone carries a smartphone, and not every phone appears in the dataset. Some people carry more than one device. Groups traveling together, such as a family in one car, a class field trip, or a hunting party, may be represented by only one or two devices, or by as many devices as there are people. Devices in airplane mode, out of battery, or without signal do not appear at all. This is one of the main reasons the dashboard is best used for comparing patterns rather than for reporting visitor counts.

Why are locations with fewer than 30 device days not displayed?

Very small samples are both unreliable and a privacy concern, so with fewer than 30 device days are suppressed.

> **MORE DETAIL** In general, a handful of observations cannot support a meaningful estimate, and reporting on a very small number of devices increases the risk that activity could be traced back to an individual. Suppressing these results protects both accuracy and privacy. A suppressed result does not mean nobody visited. It means too little was recorded to report.

4. Data Sources and Coverage

Where do the mobile location data come from?

The location data are provided by Azira, a commercial location intelligence company. The data are anonymous and are purchased under a license for this project.

> **MORE DETAIL** Azira aggregates anonymous location signals from mobile devices, sourced through third-party data partners based on users' engagement with mobile applications and advertising, where users have consented to that collection and sharing. Azira does not receive names, phone numbers, or account information, and device identifiers are cryptographically hashed. More information about Azira's data practices and privacy policy is available at www.azira.com.

How are devices included in the dataset?

Devices enter the dataset through consented data sharing associated with mobile apps and advertising. There is no way to opt into the study specifically, and no one is tracked because they visited a park.

> **MORE DETAIL** Neither CPW nor RRC selects which devices appear. The dataset is assembled by Azira from its data partners, and we receive an anonymized extract limited to Colorado. Device owners control their participation through the privacy and location settings on their own phones and in the apps they use, not through this project.

What share of recreation users does the dataset represent?

Generally in the range of 5 to 10 percent of the devices present in a study area.

> **MORE DETAIL** That share is not uniform. It can vary by location, by season, and over time, depending on cell coverage, the mix of visitors, and changes in the underlying data supply. This is precisely why the dashboard emphasizes relative comparison. A sample of roughly 5 to 10 percent is a reasonable basis for saying that one site sees more use than another, or that use grew between years, but not for stating how many people visited.

What dates are included?

All of calendar year 2024 and all of calendar year 2025.

What geographic datasets were used?

Public land boundaries come primarily from CNHP's CoMaP dataset, supplemented by other public land datasets. Roads come from Census TIGER/Line files and other roadway GIS sources. Demographic estimates come from the American Community Survey. CPW GIS data define CPW regions and properties.

MORE DETAIL **CNHP data**: the Colorado Ownership, Management, and Protection (CoMaP) public lands database provides ownership and management boundaries. COTREX provides trail data. CNHP's Conservation Summary Value dataset provides conservation context. **American Community Survey (ACS)**: used to attach demographic characteristics to inferred home locations. **Census TIGER/Line shapefiles**: used for roads, county boundaries, and Census geography. **CPW GIS data**: used for CPW regions, State Parks, State Wildlife Areas, and other CPW-managed properties.

How were study locations and their boundaries defined?

Study locations use existing agency boundaries, drawn primarily from CNHP's CoMaP data and other public land datasets. The project team did not create new boundaries.

MORE DETAIL Using existing official boundaries keeps results consistent with how agencies already describe their own lands, and makes it possible to join results to other datasets. It does mean that boundary size and shape vary a great deal between locations, which is something to keep in mind when comparing sites. See the question on location size and boundary design in the Interpreting the Results section.

How were weighted estimates calculated?

Weighted estimates combine the mobile location data sample with U.S. Census Bureau population data. These inputs help scale the sample, but the resulting Weighted Count is a modeled estimate—not a direct count of visitors.

MORE DETAIL Each public-land location was assigned to the county or counties it intersects. For each month, the population of those counties was compared with the number of device days associated with residents of the same counties. This produced a monthly multiplier representing the portion of the population captured in the mobile data sample. Because this approach relies on assumptions about sample representativeness and incorporates uncertainty from each source dataset, weighted results should be interpreted as estimates of overall magnitude rather than precise visitor totals.

How are changes in the mobile data sample over time accounted for?

The number of devices represented in the source data can vary over time for reasons unrelated to actual visitation. Results displayed in the “Results over time” section were therefore adjusted using a reliability-weighted method that places greater weight on observed results when the sample is stronger and more weight on the corresponding prior-year pattern when the sample is smaller.

MORE DETAIL For each location and week, the observed 2025 visitor-day count was evaluated relative to that location’s typical weekly 2024 sample size. The observed value was then blended with the count from the corresponding week of 2024, with limits placed on the influence of either value. A weight derived from this adjusted count was applied to the underlying 2025 visitor-day records, while 2024 served as the unadjusted baseline. This approach reduces volatility associated with changing sample sizes, but it may also smooth genuine year-to-year changes. It is separate from the Census-based weighting used to scale the mobile sample to an estimated total, and both processes produce modeled—not exact—results.

5. Methodology

What data-cleaning procedures were applied?

Raw mobile location records underwent several automated cleaning and validation procedures before analysis.

> **MORE DETAIL** When multiple coordinates were associated with the same device and timestamp, the records were deduplicated and the most complete and plausible observation was retained.
>
> Observations were also evaluated sequentially; records implying implausibly high travel speeds, based on elapsed time and straight-line distance, were excluded.
>
> To qualify as a visit, a device had to be observed within a study location for at least five minutes. This helps prevent pass-through activity and isolated detections from being counted as visits.
>
> Inferred home locations underwent separate cleaning, deduplication, and validation before being used to summarize visitor origins.
>
> These procedures improve the consistency and reliability of the results, although mobile location data remains subject to inherent limitations.

How is recreation activity distinguished from pass-through travel or other activity?

A device must be observed within a location for at least five minutes for the activity to qualify. Users can also apply roadway filters to reduce likely pass-through travel, although location data cannot determine trip purpose with certainty.

> **MORE DETAIL** Users may include activity associated with all roadways, exclude activity associated with major roads, or exclude activity associated with all mapped roads. Because travelers merely passing through public lands tend to remain along road corridors, these exclusions often provide a clearer picture of recreational use; however, they may also exclude legitimate activities such as scenic driving, cycling, or access to trailheads. Public lands can also contain residences, workplaces, utilities, and other non-recreational uses. Locations have not been individually reviewed to confirm that all observed activity is recreational, so results should be interpreted alongside maps, site characteristics, and local knowledge.

How are primary and secondary roads identified and excluded?

Roads are identified using Census TIGER/Line shapefiles together with other roadway GIS sources. Activity falling within road corridors can be excluded using the Roads filter.

Were any site-specific cleaning procedures necessary?

The statewide dashboard uses a single consistent method for every location. Site-specific adjustments are being developed as part of a separate set of case studies.

> **MORE DETAIL** Applying one consistent method statewide is what makes locations comparable to one another. Some individual sites have characteristics such as nearby residences, adjacent commercial activity, or unusual boundary shapes that would benefit from tailored handling. That work is being carried out through detailed case studies scheduled for release later this year, which will look at a smaller number of sites in greater depth.

6. Data Privacy and Anonymization

Protecting individual privacy is a requirement of this project, not an afterthought. This section explains the safeguards in place.

At what point are the data anonymized?

Before the project team ever receives them. Data are received already hashed or anonymized, in line with privacy regulations and the data provider's own policies. The data complies with all privacy regulations including California's CCPA and EU'S GDPR.

> **MORE DETAIL** RRC, CPW, and the other project partners never receive names, phone numbers, email addresses, account information, or any other direct identifier. Device identifiers are cryptographically hashed by the data provider before delivery. There is no step in the project's own workflow at which identifiable data are present, because identifiable data are never supplied and are protected in the data collection process.

Can an individual person or device be identified through the dashboard?

No. The dashboard displays only aggregated summaries of data and results based on small numbers of devices are suppressed entirely.

> **MORE DETAIL** Every number in the dashboard is a total across many devices. There is no way to select an individual device, view a single device's path, or drill down to a specific person. The minimum reporting threshold means that even in lightly used locations, no result is shown unless enough activity was recorded to keep individuals anonymous within the group.

Why are results aggregated and small samples suppressed?

Because in a small enough group, a pattern can point to a person. Aggregation and suppression together make that impossible.

> **MORE DETAIL** If a remote location recorded only two devices in a season, and one of them appeared repeatedly, someone with local knowledge might be able to guess who it was. Suppressing results below the minimum threshold removes that risk. The same practice is standard in Census reporting and public health statistics for exactly the same reason. It comes at a cost, since the most remote and lightly used places are the ones most likely to be blank, and that is a deliberate trade-off in favor of privacy.

How are visitors' inferred home locations determined?

The data provider infers a likely home area by looking at where a device consistently spends the night over a three-month period. The result is deliberately randomized so that is never more precise than a Census Block Group.

> **MORE DETAIL** Device movement is monitored by the raw data provider across roughly three months, and if a device is regularly observed at the same location at night for an average of eight or more hours, that area is treated as the device's home area. Random error, known as jitter, is then introduced so the inferred location cannot be tied to a specific household. This all happens before the project team receives the data. The most granular geography ever available to the project is the Census Block Group, an area typically containing hundreds to a few thousand people, and in the dashboard, home locations are summarized more broadly still.

How are demographics determined?

Demographics are not collected from individuals. The project team estimates them by matching a device's inferred home area to published American Community Survey data for that area.

> **MORE DETAIL** This is an area-based estimate, not a personal one. If a device's inferred home area is a Block Group where the ACS reports a certain median household income or age distribution, those area characteristics are attached to the device. It does not mean the device owner has those characteristics. It

means they live in an area that has those characteristics. Demographic results should therefore be read as broad descriptions of the communities visitors come from, not as profiles of the visitors themselves.

7. Interpreting the Results

Are the results exact visitation counts?

No. They are estimates based on a sample of roughly 5 to 10 percent of total mobile devices, and a device is not the same as a person.

> **MORE DETAIL** No number in the dashboard should be quoted as the number of people who visited a place. Even the Weighted Count, which attempts to scale the sample up, is a modeled estimate carrying real uncertainty. If you need a defensible absolute visitation figure for a specific site, treat the dashboard as a starting point and pair it with on-site counters, permit records, or a visitor survey.

Why are the data most appropriate for evaluating relative patterns, broad trends, and comparisons?

Because the sample is consistent enough to make comparisons meaningful, even though it is too incomplete to produce exact totals.

> **MORE DETAIL** The sampling limitations that make absolute counts unreliable largely cancel out when you compare. If roughly the same share of visitors is captured at two sites, the ratio between them is informative even if neither total is right. The same logic applies across time. If the sample behaves consistently from 2024 to 2025, a change in the numbers likely reflects a real change in use. This is why the dashboard is built around comparisons.

How can differences in location size or boundary design affect comparisons?

Substantially. A large area will usually record more total activity than a small one simply because it is larger, and boundaries that include highways, towns, or private inholdings can pick up activity that is not recreation.

> **MORE DETAIL** Before comparing two locations, look at the Area figure for each. Comparing a 200,000-acre national forest unit to a 40-acre state park on total device days tells you mostly about size. Boundary shape matters too. A long, narrow parcel following a highway corridor will capture a great deal of pass-through travel, while a compact parcel away from roads will not. Where sizes differ a lot, consider activity per square mile, or compare each location against its own history rather than against the other.

How should users compare years or seasons?

Compare like with like: the same location, the same season, the same metric, with the same Roads setting. Treat modest differences as noise and look for larger, sustained shifts.

> **MORE DETAIL** Two full years are available, which is enough to see a change but not enough to establish a trend. A five percent difference between 2024 and 2025 at a single site is well within the range that sampling variation alone could produce. A forty percent jump is worth investigating. Also consider what else changed. A new access road, a closure, a fire, an unusually dry winter, or a viral social media post can all move the numbers without any underlying change in an area's appeal.

What does a blank or missing result mean?

Usually that too little activity was recorded to report, meaning the result fell below the minimum threshold and was suppressed. It does not mean the location was empty.

> **MORE DETAIL** Blank results are most common at remote, lightly visited, or poorly covered locations. Occasionally a blank can also indicate that a location was not included in the underlying boundary data, or that a filter combination returned nothing. If a location you know to be well used is showing blank results, that is worth reporting to the project team.

Does zero mobile activity mean nobody visited?

No. It means no qualifying device activity was recorded.

> **MORE DETAIL** People visit public lands without phones, with phones that have no signal, with location services turned off, or with phones that simply are not part of this dataset. Areas with poor cell coverage, which describes a great deal of Colorado's backcountry, systematically under-record. Absence of data is not evidence of absence of visitors.

How can cell coverage, GPS accuracy, device settings, and observation frequency affect the results?

All four can shift results in ways that have nothing to do with actual visitation, and they tend to work against remote areas.

> **MORE DETAIL** **Cell coverage**: devices in areas with weak or no signal generate fewer records, so backcountry areas are systematically undercounted relative to front-country areas. **GPS accuracy**: accuracy varies with terrain, tree cover, and canyon walls, and a point recorded slightly off can fall on the wrong side of a boundary. **Device settings**: users who limit location sharing, use battery-saver modes, or deny app permissions contribute little or nothing. **Observation frequency**: how often a device reports depends on which apps are open and how they are configured, so two identical visits can produce very different numbers of observations. The practical implication is to be especially cautious when comparing a remote, low-coverage site to a well-covered one.

How might roads, buildings, parking areas, closures, weather, wildfire, or special events influence patterns?

Any of these can create spikes, dips, or steady background activity that looks like recreation but is not, or can suppress genuine recreation.

> **MORE DETAIL** Roads generate pass-through activity, which the Roads filter is designed to address. Buildings and facilities inside a boundary, such as administrative offices, employee housing, concessionaire operations, and private inholdings, generate activity from people who are not visitors. Parking areas concentrate signals and can make an access point look busier than the land behind it. Closures, wildfire, smoke, road washouts, and severe weather suppress use for reasons unrelated to a site's popularity. Special events such as a race, a festival, or a hunting season opener produce sharp one-time spikes. Local knowledge is essential for telling these apart, which is the subject of the next question.

Why should findings be used alongside counters, surveys, and local knowledge?

Because the dashboard tells you what happened, but rarely why, and the why is usually what a decision depends on.

> **MORE DETAIL** Each source has a different strength. Counters give accurate counts at a fixed point. Surveys explain who visitors are and what they came to do. Local staff know about the closure, the construction project, or the event that explains an odd month. The dashboard adds statewide consistency and coverage of places nothing else measures. Used together, they converge on a reliable picture. Used alone, any one of them can mislead. If the dashboard shows something that contradicts what site staff observe, the local observation is usually right, and the discrepancy is worth understanding.

8. Troubleshooting and Technical Questions

What browsers and screen sizes work best?

Any current desktop browser on a full-size screen. The dashboard works on smaller screens and tablets, but the experience is noticeably better on a desktop or laptop.

> **MORE DETAIL** The dashboard presents a lot of information at once, including maps, filters, and multiple charts, and it benefits from screen space. On phones, panels compress and some controls are harder to reach. There are no known browser incompatibilities. If a view will not load, try refreshing, and try narrowing your selection before assuming something is broken.

How do I report a suspected error?

Email the project team at movement@rrcassociates.com with a description of what you were looking at and what looked wrong.

> **MORE DETAIL** The most useful reports include the location name, the year and season selected, the metric involved, the Roads setting, a screenshot if you can take one, and a short note about what you expected to see instead. Reports from people with on-the-ground knowledge of a site are particularly valuable, because that local context is often the only way to catch a boundary or data issue.

Who should users contact with questions or requests for additional analysis?

The same address: movement@rrcassociates.com.

> **MORE DETAIL** The project team can answer questions about methodology, discuss whether the data can support a particular question, and talk through requests for custom analysis beyond what the dashboard offers. Requests involving raw data will be limited by the terms of the data license, but there is often a way to answer the underlying question.